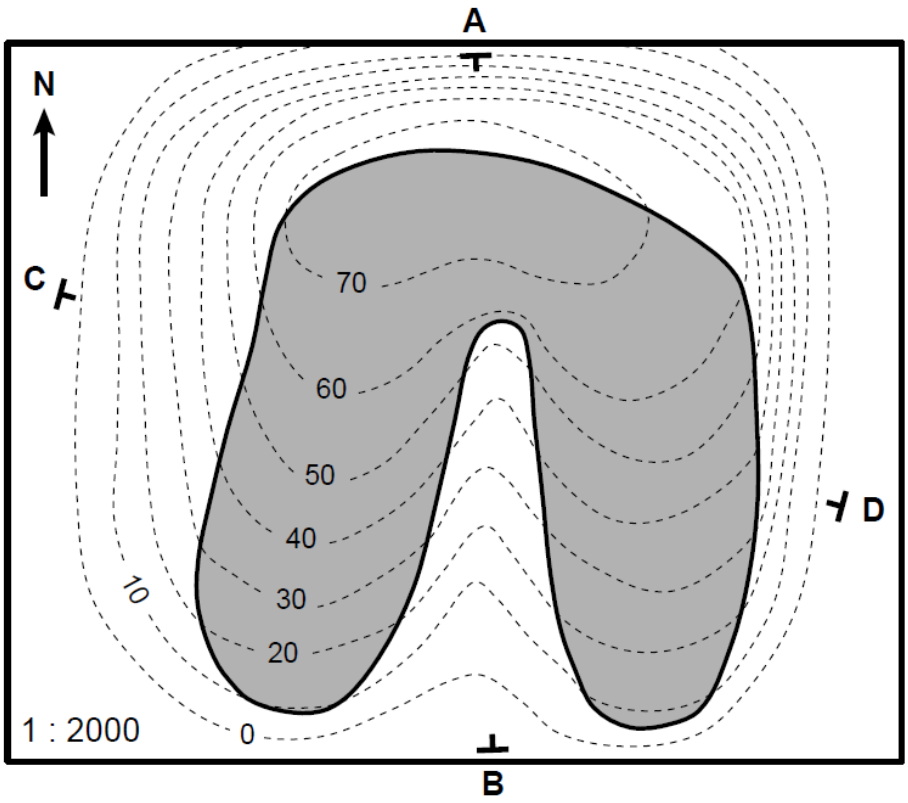
Geological Maps – 2

Determining the orientation of surfaces from outcrop patterns

The map below shows a hillside on which there are some mudstones which are overlain by a sandstone unit (shaded). The topographic contours are in metres.

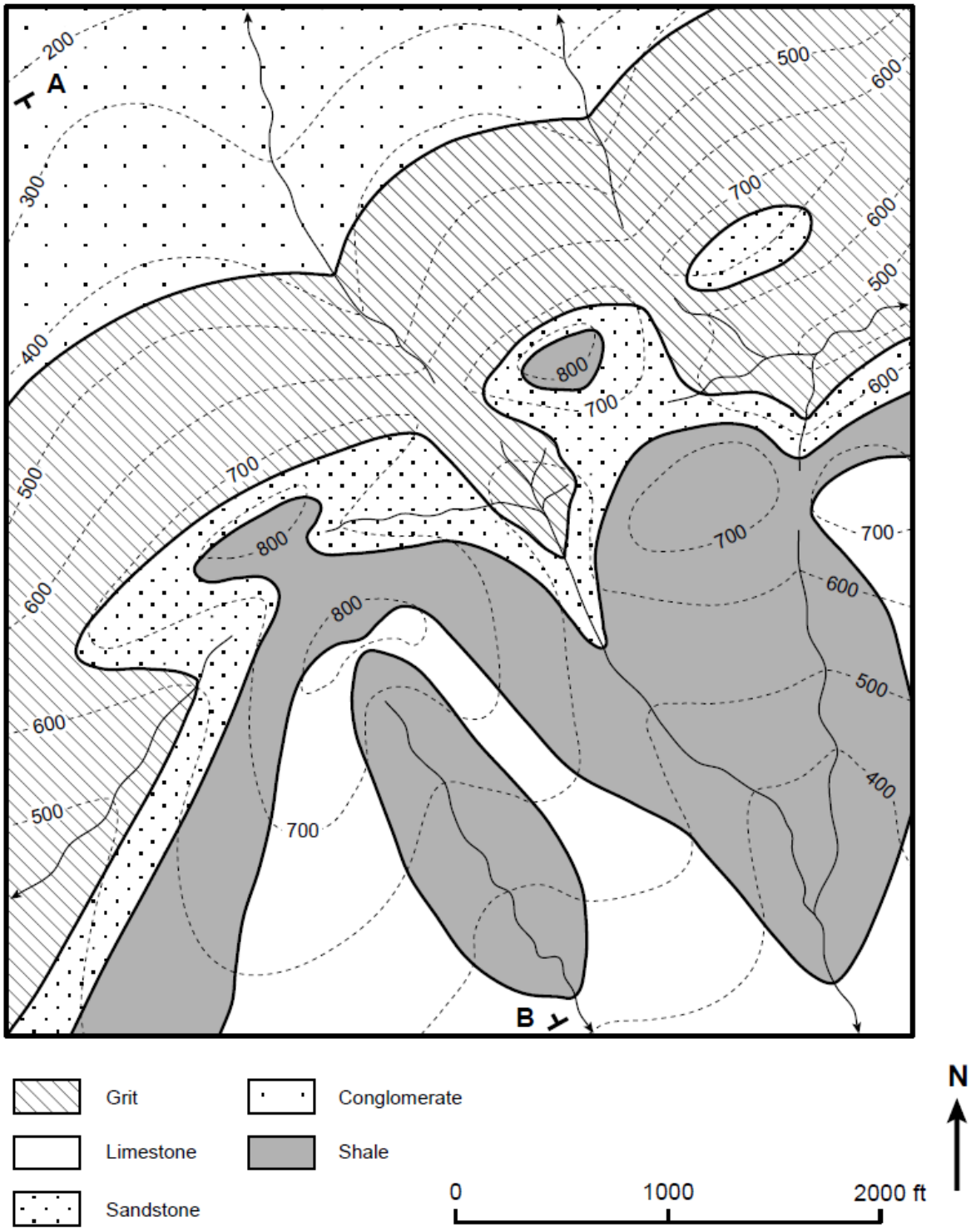
Exercise 1

(a) Construct stratum contours for the contact between the mudstone and sandstone, labelling each with its height above sea level.
(b) Determine the strike, dip and dip direction of the contact.
(c) Draw a vertical cross-section from A-B showing the topography and contact
(d) Repeat (c) for the line along C-D.



Drawing a cross-section using stratum contours

The map below shows a uniformly dipping sequence of sediments.



Exercise 2

(a) Draw stratum contours on the contact between the shale and limestone and determine the strike, dip and dip direction of the sediments.

(b) Using the graph paper provided, draw a vertical cross-section along A-B.

True thicknesses of dipping beds

The true thickness of a bed is the distance between the top and bottom, in a direction perpendicular to the stratification. In vertical sections, dipping beds exaggerate the to some larger value (*V*). Figure 1 shows the difference between the true thickness (*t*), the vertical thickness (*V*) and the outcrop width (*w*) and give equations for calculating *t* if the dip is known.

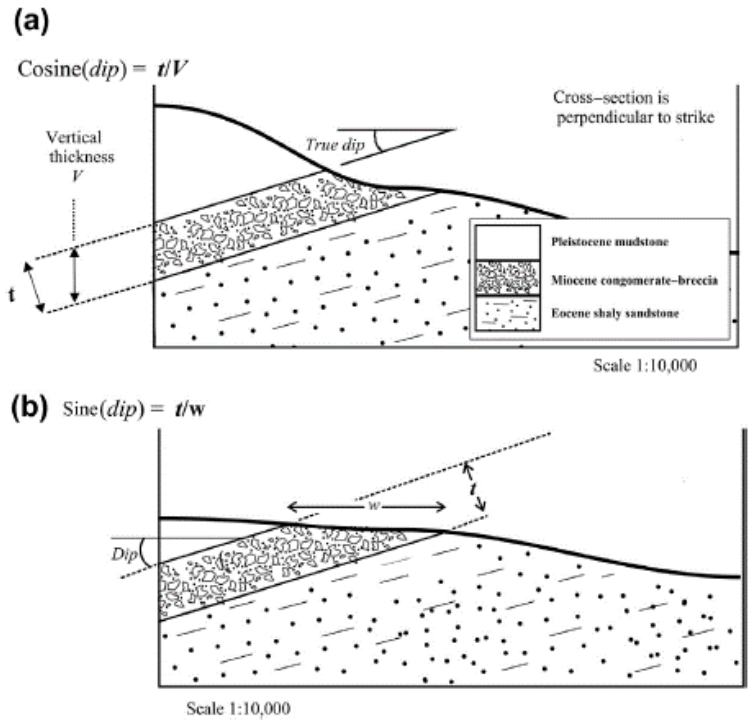


Figure 1. Determining true (*t*) thickness of strata. Using the equations shown: (a) if the vertical thickness (*V*) and the dip are known then the true thickness (*t*)can be calculated; and (b) if the outcrop width (*w*) and dip are known then *t* can be calculated.

(c) Use your cross-section to construct a stratigraphic column showing the sedimentary units in correct order and their true thicknesses. Draw this on your graph paper next to the cross-section.

Drawing a cross-section on a map with an unconformity

So far we have drawn stratum contours on maps where all the strata of different ages have the same orientation (e.g. Exercise 2). In reality, different series of strata may have different orientations (i.e. have different strikes and dips), this is always the case if there is an angular unconformity present on the map. NB remember from your EART11100 Understanding the Earth workshops that an angular unconformity is where older strata are tilted, eroded and exposed as a depositional surface and then an unconformable sequence of younger strata are deposited on top of them.

Exercise 3

The map shown overleaf shows two series of strata: E and F overlie C and D unconformably. The unconformity is at the base of E.

(1) Identify and label the unconformity on the map.

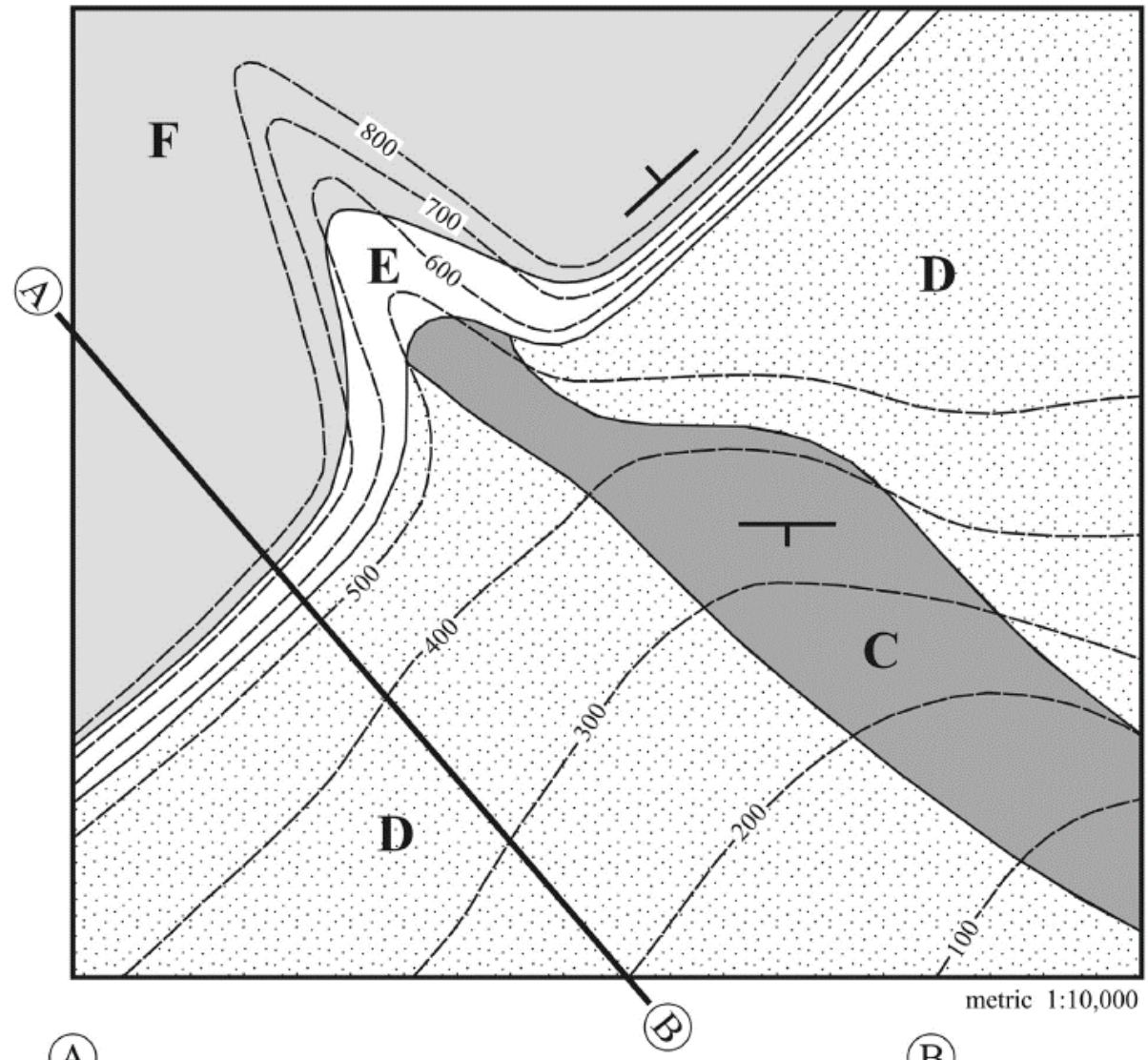
(2) What is the main topographic feature indicated by the contours?

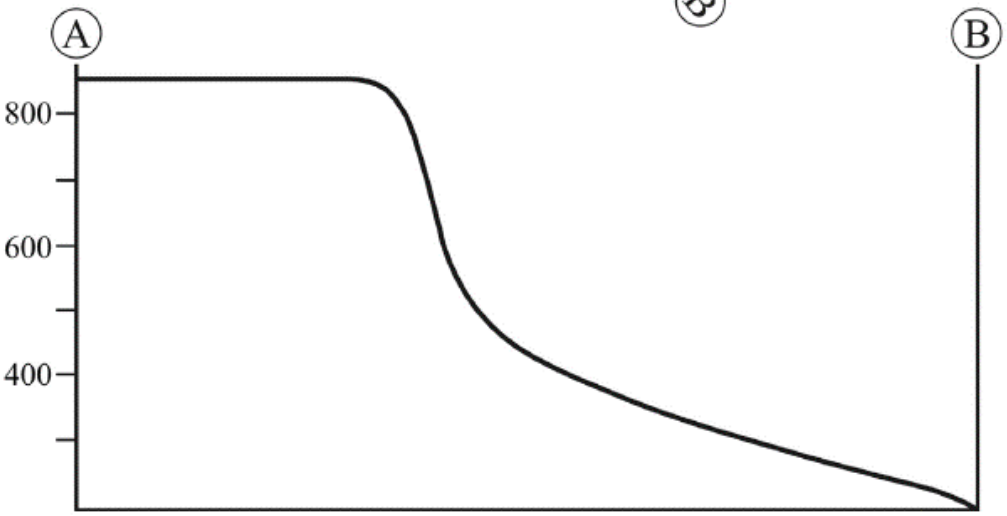
(3) What is the dip direction that you infer from the shape of the contact for E and F?

(4) Find the bedding strike/dip symbol on the map for C, what is the approximate orientation of the strike (use N-S, NE-SW, E-W etc), what is the direction of dip?

(5) Draw a vertical cross-section along the line A-B.
    a. Start by drawing stratum contours on the contact between E-F (assume a plane-dipping surface) and mark this contact on your cross-section.
    b. Now draw stratum contours for the contact between C and D and "project" these onto your cross-section. Do this by extending the stratum contours to the A-B line and then marks points at the corresponding positions and heights on your cross-section. This will allow you to draw the C-D contact as a line on your section.
    c. Complete your cross-section by labelling each stratum with its letter code, and ornamentation.

(6) Give the strike/dip and dip direction of strata E and F and C and D.





Some of the maps and diagrams used in this practical are from Borradaile G. (2014) Understanding Geology Through Maps. First Edition, Elsevier, 183 pp.